SIMATS ENGINEERING

ASSESSMENT TOOL 2: SCENARIO-BASED

COURSE NAME: COMPUTER VISION with OpenCV
COURSE CODE: DSA0204

COURSE OUTCOME COVERED

Assessment Tool Weightage: 20%

Name: V M Sai Bhargav

<u>Scenario-Based Questions — Answers</u>

1. Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

(a) Low contrast occurs when the pixel intensity values of an image are clustered within a narrow band of the total available range (0–255), rather than being spread across the full range. As a result, the object and background occupy nearly the same gray levels, and the human eye — or any downstream algorithm — struggles to distinguish one region from another. This is typically caused by poor lighting conditions during image capture, a limited dynamic range of the imaging sensor, or improper exposure settings that compress the scene's true intensity variation into a small digital range.

(b) The key issues affecting visibility are: (i) a histogram that is narrow and concentrated around a small range of gray levels; (ii) very small intensity differences between the object and background, making edges and details hard to perceive; (iii) loss of subtle structural information because fine variations are compressed; and (iv) reduced effectiveness of any subsequent processing (e.g., segmentation or edge detection) that relies on intensity differences.

(c) Histogram equalization is applied to redistribute the pixel intensities so that they span the entire available dynamic range. This is done by computing the cumulative distribution function (CDF) of the image histogram and using it as a transformation function to remap the original intensities to new, more widely spread values. Contrast stretching (linear scaling of intensities to the full range) is a simpler alternative for milder cases.

(d) Histogram equalization is preferred because it is a fully automatic, global technique that does not require manual parameter tuning — it directly analyzes the image's own histogram and produces a transformation that maximizes the spread of intensity values. Unlike a simple linear stretch, it accounts for the actual distribution of pixel values, giving more contrast enhancement to the most populated intensity regions. For images where different local areas require different degrees of enhancement, Contrast Limited Adaptive Histogram Equalization (CLAHE) would be an even more suitable refinement, as it applies equalization locally while limiting noise amplification.

(e) Scenario → Cause → Key Issues → Technique → Justification, presented in that logical sequence, gives a clear and structured explanation that mirrors the diagnostic process an image-processing engineer would follow in practice.

2. Uneven Illumination

An image captured under non-uniform lighting shows bright and dark patches.

(a) Uneven illumination means the recorded intensity at each pixel depends not only on the true reflectance/albedo of the object but also on the spatially varying amount of incident light falling on the scene. Regions closer to the light source, or oriented favorably towards it, appear artificially bright, while regions in shadow or facing away appear artificially dark — even if the underlying material is uniform. This distorts the true appearance of the scene and can mislead any analysis based on absolute intensity values.

(b) Key factors causing this intensity variation include: (i) a single directional or point light source rather than diffuse, uniform lighting; (ii) cast shadows from objects or the camera setup; (iii) sensor vignetting, where corners of the image receive less light than the center; and (iv) uneven surface reflectance/orientation relative to the light source.

(c) Homomorphic filtering is applied to normalize illumination. The image is modeled as the product of an illumination component (slowly varying, low frequency) and a reflectance component (rapidly varying, high frequency). Taking the logarithm converts this multiplicative model into an additive one, after which a high-pass filter in the frequency domain suppresses the illumination component while preserving the reflectance (detail) component; the exponential is then taken to return to the spatial domain. Adaptive histogram equalization (CLAHE) is a suitable simpler alternative.

(d) Homomorphic filtering is justified because it directly targets the physical cause of the problem — it separates illumination from reflectance based on their distinct frequency characteristics, allowing the unwanted low-frequency illumination variation to be suppressed without destroying the high-frequency reflectance detail that carries the true object information. This is more principled than a simple global contrast adjustment, which cannot correct spatially varying illumination.

(e) Presenting the answer as Cause → Contributing Factors → Method → Reasoning keeps the explanation logically structured and easy to follow.

3. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

(a) Salt-and-pepper noise (also called impulse noise) manifests as randomly scattered pure white (salt) and pure black (pepper) pixels superimposed on the image. Unlike Gaussian noise, it does not affect every pixel by a small amount; instead, a small fraction of pixels are corrupted entirely, taking on extreme intensity values regardless of their true value. In a scanned document, this typically arises from dust, scanner sensor faults, or transmission errors, and it severely disrupts the readability of text by introducing spurious dark or light specks.

(b) Key issues affecting quality include: (i) isolated, extreme-valued pixels that do not correlate with neighboring pixels; (ii) obscured or broken characters that reduce readability and OCR accuracy; and (iii) the impulsive, bipolar nature of the corruption, which makes linear smoothing filters ineffective and can even spread the noise.

(c) A median filter is applied. For each pixel, a neighborhood window (e.g., 3×3) is considered, and the pixel value is replaced with the median of the intensities in that neighborhood.

(d) The median filter is more suitable than a mean/averaging filter because the median is a non-linear, rank-based statistic that is insensitive to extreme outlier values — the salt and pepper pixels are simply outvoted by the surrounding normal pixels and excluded from the output, whereas a mean filter would average the extreme values into neighboring pixels, spreading rather than removing the noise and also blurring the image.

(e) Explaining noise type → impact → chosen filter → comparison with alternatives gives a clear, logical progression.

4. Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

(a) When a smoothing filter (such as a mean or Gaussian filter) is applied to remove noise, it averages each pixel with its neighbors. While this suppresses random noise variations effectively, it also averages across true intensity discontinuities — i.e., edges — because the filter cannot distinguish between noise fluctuations and genuine sharp transitions. The result is a softened, blurred transition at object boundaries instead of a crisp edge.

(b) Key issues are: (i) loss of edge sharpness and reduced distinctness of object boundaries; (ii) an inherent trade-off between the degree of noise removal and the amount of detail/edge information preserved; and (iii) larger filter kernels producing more noise suppression but proportionally more edge blurring.

(c) Unsharp masking is applied to restore edge sharpness: a blurred version of the image is subtracted from the original to obtain a high-frequency detail (edge) mask, which is then added back (scaled by a factor) to the original image, boosting the edge transitions. Alternatively, an edge-preserving smoothing filter such as the bilateral filter could have been used originally to avoid the blur in the first place.

(d) Unsharp masking is justified because it operates directly on the high-frequency component that was lost, re-injecting edge contrast without needing to redo the entire noise-removal process; it selectively enhances transitions rather than uniformly sharpening the whole image, which limits unwanted noise amplification compared to a naive high-pass boost.

(e) A logical flow of Cause of blur → Consequences → Restoration method → Reasoning presents the solution clearly.

5. High-Frequency Noise

A satellite image contains fine-grained noise affecting feature visibility.

(a) High-frequency noise refers to rapid, small-scale pixel-to-pixel intensity fluctuations that appear as fine graininess throughout the image. In the frequency (Fourier) domain, this noise corresponds to energy concentrated at high spatial frequencies, distinct from the generally lower-frequency content of meaningful large-scale features such as terrain, water bodies, or structures.

(b) Key issues affecting clarity include: (i) fine noise visually competing with genuine fine-scale features (e.g., small structures), making them hard to distinguish; (ii) reduced signal-to-noise ratio at high frequencies; and (iii) the difficulty of removing this noise in the spatial domain without also blurring genuine fine detail.

(c) A frequency-domain low-pass filter is applied: the image is transformed via the Fast Fourier Transform (FFT), a low-pass filter (e.g., a Butterworth or Gaussian low-pass filter) is applied to attenuate high-frequency components, and the inverse FFT is taken to recover the denoised spatial-domain image.

(d) A Butterworth (or Gaussian) low-pass filter is preferred over an ideal low-pass filter because the ideal filter has a sharp cutoff in the frequency domain, which produces ringing artifacts (Gibbs phenomenon) in the spatial domain. The smooth, gradual roll-off of the Butterworth/Gaussian filter avoids these artifacts while still effectively suppressing the high-frequency noise energy.

(e) The response follows Definition of noise → Impact → Frequency-domain method → Justification, in a clear, organized manner.

6. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

(a) Accurate boundary detection is critical in medical imaging because it delineates the precise extent of anatomical structures, organs, or abnormalities such as tumors. Clinically, this directly affects measurements (e.g., tumor size), diagnosis, and treatment planning, so the edges must be both accurately localized and continuous.

(b) Key challenges include: (i) low intensity contrast between adjacent soft tissues; (ii) noise inherent to medical imaging modalities (e.g., ultrasound speckle, MRI thermal noise); and (iii) blurred or partially broken boundaries caused by the imaging process itself.

(c) The Canny edge detector is applied. It involves Gaussian smoothing to reduce noise, computation of intensity gradients (magnitude and direction), non-maximum suppression to thin edges to single-pixel width, and hysteresis thresholding (using two thresholds) to link and retain only genuinely strong, connected edges.

(d) Canny is selected over simpler operators (e.g., Sobel or Prewitt) because it combines noise suppression, precise localization, and a single, thin response per true edge through non-maximum suppression, and its hysteresis thresholding recovers weak-but-connected edge segments that a single-threshold method would discard — all of which are essential for reliably tracing subtle tissue boundaries.

(e) Presenting Importance → Challenges → Method → Justification keeps the medical reasoning systematic and clear.

7. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

(a) Since the object and background occupy clearly separated ranges of intensity, the segmentation requirement here is straightforward: a single dividing intensity value (threshold) can be found that correctly classifies each pixel as either object or background.

(b) The key feature enabling this separation is a bimodal histogram — two distinct peaks corresponding to the object and background intensity populations, with a clear valley between them where a threshold can be placed.

(c) Global thresholding using Otsu's method is applied. Otsu's algorithm analyzes the histogram and automatically selects the threshold that minimizes the within-class variance (equivalently, maximizes the between-class variance) of the two resulting pixel classes.

(d) Otsu's method is justified because it removes the need for manual threshold selection, works optimally when the histogram is bimodal (as it is here), and is computationally efficient, making it the natural and reliable choice for this simple, well-separated case.

(e) The explanation follows Requirement → Enabling feature → Method → Reasoning in a concise, clear structure.

8. Complex Segmentation

An image contains multiple regions with overlapping intensity values.

(a) When multiple regions share overlapping intensity ranges, a single global threshold cannot cleanly separate them, since some pixels from different true regions will fall on the same side of any chosen threshold. This makes segmentation considerably more challenging than the simple bimodal case.

(b) Key issues include: (i) a histogram that is not clearly multimodal, with overlapping peaks; (ii) ambiguity in assigning pixels near the overlap region to the correct class; and (iii) the failure of any single-threshold approach to correctly separate all regions simultaneously.

(c) An advanced segmentation method such as k-means clustering (or watershed segmentation for touching regions) is applied. K-means groups pixels into k clusters based on similarity in a feature space (e.g., intensity, and optionally spatial coordinates or color), iteratively refining cluster assignments and centroids until convergence.

(d) K-means (or watershed) handles the overlap by using more information than a single intensity value — it considers the overall distribution and relative distances of pixels in feature space, or, in the case of watershed, the topology of the intensity gradient — allowing it to resolve ambiguous, overlapping regions that a simple threshold cannot.

(e) The response follows Challenge → Contributing issue → Advanced method → How it resolves complexity, presented clearly.

9. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

(a) Broken edges occur when parts of an object's true boundary produce a weak or inconsistent gradient response — due to noise, low local contrast, or aggressive thresholding during edge detection — causing those segments to be missed while other parts of the same boundary are detected, resulting in a discontinuous outline.

(b) Key issues affecting boundary continuity include: (i) noise disrupting the gradient computation at certain points; (ii) locally weak intensity transitions along parts of the true boundary; and (iii) a thresholding step that removes genuinely weak but valid edge pixels.

(c) Morphological closing (dilation followed by erosion) is applied to bridge small gaps in the edge map, or alternatively edge-linking techniques such as the Hough transform can be used to connect edge fragments that lie along a common line/curve.

(d) Morphological closing is justified because dilation first expands edge segments enough to bridge small gaps, and the subsequent erosion restores the original edge thickness, so the overall shape and size of the boundary are preserved while continuity is restored — making it a simple, effective, and shape-preserving solution to small discontinuities.

(e) The explanation proceeds Cause of breakage → Effect → Reconnection method → Reasoning, presented clearly.

10. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

(a) This scenario requires a combined objective: noise present in the captured image must be suppressed so it does not interfere with automated defect detection, while at the same time the edges corresponding to true defects (cracks, scratches, etc.) must remain sharp and well-defined, since these edges are exactly what the inspection system needs to detect.

(b) Key issues include: (i) the inherent trade-off where standard smoothing filters that remove noise also blur the defect edges; and (ii) the risk that if edges are preserved too aggressively, residual noise may be mistaken for defects (false positives).

(c) A bilateral filter is applied, which smooths the image using a combination of spatial proximity and intensity similarity weighting, so that noise in relatively homogeneous regions is averaged out while strong intensity discontinuities (edges) are left largely intact. Alternatively, a sequential pipeline of median filtering followed by Canny edge detection can be used.

(d) The bilateral filter (or the denoise-then-detect pipeline) is justified because it directly addresses both requirements in a principled way: the range (intensity) weighting term specifically prevents averaging across strong edges, so noise removal does not come at the cost of losing the boundaries needed for defect detection. When using a pipeline, denoising is performed before edge detection so that noise-induced gradients are not mistaken for true defect boundaries.

(e) Presenting Combined requirement → Trade-off → Chosen method → Reasoning for order/selection gives a clear, structured answer.

11. Low Brightness Image

An image captured in low light appears very dark with poor visibility.

(a) Low brightness occurs because insufficient light reached the sensor during capture, causing pixel intensities to be concentrated in the lower (dark) end of the available range. This compresses the true scene detail into a small band of low intensity values, making it difficult to perceive objects clearly.

(b) Key issues affecting visibility include: (i) intensities clustered near zero, underutilizing the available dynamic range; (ii) loss of perceptible detail in shadowed/dark regions; and (iii) low signal-to-noise ratio, since sensor noise becomes proportionally more significant at low light levels.

(c) Gamma correction with a gamma value less than 1 ($\gamma < 1$) is applied, which non-linearly expands the darker intensity range, brightening the image while compressing the already-bright region less aggressively. A logarithmic transformation is a related alternative.

(d) Gamma correction ($\gamma < 1$) is preferred over simply adding a constant brightness offset because it non-linearly expands dark tones more than bright tones, mimicking human perceptual sensitivity and revealing detail in shadows without saturating or excessively brightening the parts of the image that were already reasonably visible.

(e) The explanation proceeds Cause → Issues → Method → Reasoning in a clear, structured manner.

12. Overexposed Image

An image appears excessively bright, losing important details.

(a) Overexposure occurs when too much light reaches the sensor, causing many pixel values to saturate at or near the maximum intensity (255 for 8-bit images). Once saturated, distinct real-world intensity differences in bright regions collapse to the same maximum value, so detail in those areas is irrecoverably lost in the captured data, though correction can still improve visibility of the remaining non-saturated information.

(b) Key issues include: (i) clipped highlights where multiple true intensity levels map to the same maximum pixel value; (ii) loss of information/detail in bright regions; and (iii) reduced effective dynamic range being used for meaningful scene content.

(c) Gamma correction with a gamma value greater than 1 ($\gamma > 1$) is applied to compress the bright-intensity range, redistributing values so the non-saturated bright details become more distinguishable. Histogram equalization is a viable alternative.

(d) Gamma correction with $\gamma > 1$ is justified because it selectively compresses the higher end of the intensity scale more than the lower end, which helps recover and better distinguish the remaining detail near saturation, whereas a simple linear brightness reduction would darken the entire image uniformly without specifically addressing the compressed highlight information.

(e) The response follows Cause of overexposure → Issues → Correction technique → Reasoning, presented in a structured manner.

13. Gaussian Noise

An image contains Gaussian noise affecting smooth regions.

(a) Gaussian noise is a form of additive noise where each pixel's value is corrupted by a random value drawn from a Gaussian (normal) probability distribution, independent of position. It appears as a fine, grainy variation superimposed uniformly across the image, and is most visually noticeable in otherwise smooth, flat-intensity regions where the true value should be constant.

(b) Key issues affecting quality include: (i) increased local variance in what should be homogeneous regions; (ii) reduced overall image clarity and a grainy appearance; and (iii) the fact that this noise affects every pixel to some degree, unlike sparse impulse noise.

(c) A Gaussian smoothing filter (a weighted averaging filter using a Gaussian kernel) is applied to reduce the noise by averaging each pixel with its neighbors, weighted more heavily towards the center.

(d) The Gaussian filter is preferred over a plain averaging (box) filter because its weighting matches the statistical, bell-shaped nature of the noise distribution and gives more importance to nearby pixels than distant ones, resulting in effective noise reduction with comparatively less blurring than a uniform box filter of the same size. For further optimality in the mean-square-error sense, a Wiener filter could be used as it explicitly models the noise statistics.

(e) The response follows Noise description → Issues → Filter choice → Comparative reasoning, presented clearly.

14. Speckle Noise

An image obtained from ultrasound imaging shows speckle noise.

(a) Speckle noise is a granular, multiplicative form of noise (rather than additive) that arises from the constructive and destructive interference of coherent ultrasound wave reflections from tissue microstructures. It manifests as a grainy, salt-like texture whose intensity is proportional to the underlying signal, rather than being independent of it.

(b) Key issues affecting clarity include: (i) its multiplicative nature, meaning the noise magnitude scales with the local signal intensity; (ii) reduced contrast and obscured fine tissue structures; and (iii) the fact that filters designed for additive noise are not optimally suited to it.

(c) A speckle-specific filter such as the Lee filter (or an anisotropic diffusion filter) is applied, which adapts its smoothing based on local statistics (mean and variance) computed within a moving window.

(d) The Lee filter is justified because it explicitly models the multiplicative noise process and adapts its filtering strength locally — smoothing more in homogeneous areas (where speckle dominates) and less near edges/structures (where signal variance is genuinely high) — providing better speckle suppression with less structural detail loss than a generic filter designed for additive Gaussian noise.

(e) The response follows Noise characteristics → Issues → Filter selection → Reasoning, presented clearly.

15. Loss of Fine Details

An image loses fine details after processing.

(a) Loss of fine detail typically occurs as a side effect of noise-removal or smoothing operations: since fine details (like texture or small structures) and noise both correspond to high-frequency content, an aggressive smoothing filter suppresses both indiscriminately, removing genuine detail along with the unwanted noise.

(b) Key issues related to filtering include: (i) the inherent overlap in frequency content between true fine detail and noise, making them hard to separate; (ii) larger filter kernel sizes or higher smoothing strength producing greater detail loss; and (iii) the general trade-off between noise suppression and detail preservation.

(c) An edge-preserving/detail-preserving technique such as the bilateral filter is applied, or unsharp masking is used after denoising to reintroduce lost high-frequency detail.

(d) The bilateral filter is justified because its range (intensity-similarity) weighting term ensures that smoothing only occurs among pixels with similar intensity, which naturally preserves genuine fine structures and edges (which have distinct neighboring intensities) while still averaging out random noise fluctuations in truly homogeneous areas.

(e) The response follows Cause of detail loss → Issues → Remedial technique → Reasoning, presented clearly.

16. Edge Enhancement Requirement

An application requires highlighting edges for feature extraction.

(a) Certain applications (e.g., object recognition, feature-based matching) rely on strong, clear edge information as the primary input feature; if edges are weak or subtle, the subsequent feature-extraction step will be unreliable, so the edges must first be deliberately enhanced/highlighted.

(b) Key issues in detecting edges include: (i) inherently low contrast at some true boundaries; (ii) noise partially masking the intensity gradient at edges; and (iii) the need to boost genuine edge signal without amplifying noise disproportionately.

(c) Laplacian sharpening (or unsharp masking) is applied: the Laplacian operator, which responds strongly to regions of rapid intensity change, is computed and added back to the original image to accentuate the edges.

(d) The Laplacian-based approach is justified because the Laplacian, being a second-derivative operator, has a strong, zero-crossing response precisely at edges, and combining it with the original image amplifies exactly those transition regions, producing a visibly edge-enhanced result suitable for subsequent feature extraction.

(e) The response follows Requirement → Detection issues → Enhancement method → Reasoning, presented clearly.

17. Background Noise Removal

An image contains background noise affecting segmentation accuracy.

(a) Noise present in the image introduces spurious intensity variations that do not correspond to true scene structure; when segmentation is performed directly on such an image, these variations can be misinterpreted as genuine boundaries or separate regions, degrading segmentation accuracy.

(b) Key issues affecting segmentation include: (i) noise-induced false edges or false region boundaries; (ii) fragmentation of what should be a single homogeneous region into many small, noise-driven sub-regions; and (iii) reduced reliability of any intensity-based grouping criterion.

(c) A suitable preprocessing filter — such as a Gaussian or median filter — is applied to the image before segmentation, to suppress noise while retaining the essential structural information needed for accurate region identification.

(d) Pre-filtering is justified because segmentation algorithms assume that intensity variation reflects true scene structure; removing noise beforehand ensures that the segmentation step operates on a cleaner signal, directly improving the accuracy and coherence of the resulting regions.

(e) The response follows Noise impact → Segmentation issues → Preprocessing method → Reasoning, presented clearly.

18. Threshold Selection Problem

An image requires segmentation, but selecting a threshold is difficult.

(a) When an image's histogram does not have a clear, well-separated bimodal shape, or when illumination varies across the image, a single manually or globally chosen threshold value will not correctly separate object from background everywhere, making threshold selection genuinely difficult.

(b) Key issues in threshold selection include: (i) subjectivity and error-proneness of manual threshold choice; (ii) illumination variation across the image causing the appropriate threshold to differ from one region to another; and (iii) an ambiguous or absent valley in the histogram.

(c) Adaptive (local) thresholding is applied, where a different threshold is computed for each pixel or small region based on the local neighborhood statistics (e.g., local mean or Gaussian-weighted mean), rather than a single global value.

(d) Adaptive thresholding is justified because it directly compensates for spatial illumination variation — by tailoring the threshold to local conditions, it correctly segments objects even when a single global threshold (such as one chosen via Otsu's method) would fail across the whole image.

(e) The response follows Challenge → Selection issues → Thresholding method → Reasoning, presented clearly.

19. Texture-Based Segmentation

An image contains regions distinguishable by texture rather than intensity.

(a) In this scenario, different regions may share very similar average intensity values but differ in their spatial pattern of intensity variation (i.e., texture, such as smooth versus rough or patterned regions), meaning that intensity-based segmentation methods have no basis for distinguishing them.

(b) Key issues with intensity-based methods include: (i) similar or overlapping mean intensities across texturally distinct regions; (ii) thresholding or clustering on intensity alone producing meaningless or incorrect region boundaries; and (iii) the loss of spatial pattern information when only per-pixel intensity is considered.

(c) A texture-based segmentation approach is applied, using texture descriptors such as Local Binary Patterns (LBP), Gabor filter responses, or Gray-Level Co-occurrence Matrix (GLCM) features, followed by clustering or classification based on these descriptors rather than raw intensity.

(d) This approach is justified because texture descriptors explicitly capture the spatial arrangement and statistical pattern of intensity variation within a neighborhood (e.g., directionality, frequency, coarseness), which is precisely the property that differs between the regions — something a single-pixel intensity value cannot represent.

(e) The response follows Challenge → Intensity-method limitation → Texture-based approach → Reasoning, presented clearly.

20. Multiple Object Detection

An image contains multiple objects with similar intensity values.

(a) When several distinct objects share similar intensity levels, a segmentation method relying purely on intensity thresholding cannot separate them from one another, even though they are spatially distinct and possibly touching or close together, because the algorithm has no way to distinguish 'object A' from 'object B' using intensity alone.

(b) Key issues in distinguishing objects include: (i) identical or very close intensity values across different true objects; (ii) potential touching or overlapping of objects, complicating boundary identification; and (iii) the need for spatial/shape cues in addition to, or instead of, intensity.

(c) Watershed segmentation (often combined with a distance transform to identify markers) is applied. The distance transform of the thresholded foreground identifies local intensity peaks corresponding to individual object centers, which serve as markers/seeds; the watershed algorithm then 'floods' from these markers to delineate each separate object.

(d) Marker-controlled watershed segmentation is justified because it uses spatial/topological information (the distance transform reveals separate object cores even when objects touch) rather than relying solely on intensity differences between objects, making it capable of separating multiple similar-intensity objects that a simple threshold would merge into one region.

(e) The response follows Difficulty → Distinguishing issues → Segmentation technique → Reasoning, presented clearly.

21. Edge Noise Problem

Edges detected are affected by noise.

(a) When noise is present in an image, it introduces small, random intensity fluctuations that create spurious local gradients; a naive edge detector responds to these fluctuations just as it would to a true edge, resulting in an edge map cluttered with false edges alongside (or instead of) the genuine object boundaries.

(b) Key challenges in edge detection under noise include: (i) false positive edges arising purely from noise gradients; (ii) genuine but weak edges being obscured or indistinguishable from noise-induced gradients; and (iii) the sensitivity of simple gradient operators (e.g., Sobel) to high-frequency noise.

(c) The Canny edge detector is applied, which begins with Gaussian smoothing of the image to suppress noise before gradient computation, followed by non-maximum suppression and hysteresis (dual) thresholding to retain only strong, well-connected true edges.

(d) Canny is justified because its initial Gaussian smoothing step specifically attenuates the high-frequency noise that would otherwise generate false gradients, and its hysteresis thresholding distinguishes genuinely significant edges (and weak edges connected to them) from isolated noise-driven responses, which a single-threshold gradient method cannot do as reliably.

(e) The response follows Problem → Detection challenges → Solution → Reasoning, presented clearly.

22. Region Merging Requirement

An image contains fragmented regions that need merging.

(a) Fragmentation occurs when a segmentation process (e.g., one sensitive to noise or fine intensity variation) divides what should be a single coherent object or area into many small, disconnected sub-regions, none of which individually represents the true, meaningful structure.

(b) Key challenges in region segmentation here include: (i) determining a similarity criterion by which adjacent small regions should or should not be merged; (ii) avoiding both under-merging (leaving fragments unmerged) and over-merging (combining genuinely distinct regions); and (iii) computational considerations when many small regions are present.

(c) A split-and-merge algorithm (or region-growing-based merging) is applied: adjacent regions are compared using a homogeneity/similarity criterion (e.g., similar mean intensity or texture), and regions satisfying this criterion are merged into a single, larger, coherent region.

(d) This approach is justified because it uses an explicit, quantitative similarity criterion to decide whether adjacent fragments genuinely belong together, systematically producing coherent regions from noise-driven fragments while stopping short of merging genuinely distinct structures that fail the similarity test.

(e) The response follows Fragmentation issue → Merging challenges → Technique → Reasoning, presented clearly.

23. Over-Segmentation

An image is divided into too many regions after segmentation.

(a) Over-segmentation occurs when the segmentation algorithm is overly sensitive to minor intensity or gradient variations (often caused by noise or texture), producing a much larger number of regions than there are true, semantically meaningful objects or areas in the scene — a common failure mode of, for example, naive watershed segmentation applied directly to a noisy gradient image.

(b) Key issues affecting segmentation here include: (i) noise or texture creating many spurious local minima/maxima that each spawn a separate region; (ii) loss of meaningful, larger-scale grouping; and (iii) increased computational and interpretive burden from handling excessive regions.

(c) Marker-controlled watershed segmentation (or a post-processing region-merging step) is applied, restricting the flooding process to a limited, deliberately chosen set of markers corresponding to true objects of interest, rather than allowing every local minimum to become a separate region.

(d) Marker-controlled watershed is justified because it constrains segmentation to meaningful seed points (determined, for instance, via smoothing and local-maxima detection), directly suppressing the excessive, noise-driven fragmentation that plain watershed produces, thereby correcting the over-segmentation.

(e) The response follows Description of over-segmentation → Contributing issues → Correction method → Reasoning, presented clearly.

24. Under-Segmentation

Important regions are merged incorrectly during segmentation.

(a) Under-segmentation is the opposite failure mode: the algorithm fails to detect a sufficiently strong boundary between two genuinely distinct regions (e.g., due to weak contrast or an overly aggressive merging criterion), causing them to be incorrectly grouped together as if they were a single region.

(b) Key issues affecting separation include: (i) weak or gradual intensity transitions between adjacent true regions; (ii) an insufficiently strict merging/threshold criterion; and (iii) inadequate marker/seed placement when using region-growing or watershed methods, causing one seed to dominate multiple true regions.

(c) Improved marker selection using the distance transform (to place one distinct seed per true object) combined with marker-controlled watershed segmentation is applied, ensuring that each genuinely separate region receives its own seed and is not absorbed into a neighboring region.

(d) This approach is justified because explicitly ensuring one marker per true object — rather than relying on a global criterion that might treat weakly-separated regions as one — directly targets the root cause of under-segmentation, forcing the algorithm to respect the true number and location of distinct regions.

(e) The response follows Description → Separation issues → Solution → Reasoning, presented clearly.

25. Boundary Refinement

Detected boundaries are rough and inaccurate.

(a) Rough, inaccurate boundaries typically result from an initial detection process (e.g., simple thresholding or basic edge detection) that is sensitive to noise and local intensity fluctuations, producing a jagged contour that only approximately follows the true, smooth object boundary rather than tracing it precisely.

(b) Key challenges in boundary detection here include: (i) noise causing local deviations along the detected contour; (ii) the initial detection method lacking any global shape or smoothness constraint; and (iii) the need to refine the boundary without losing genuine shape features.

(c) An active contour model (snake) is applied: starting from the initial rough boundary, the contour is iteratively deformed to minimize an energy function that balances internal smoothness/continuity terms with an external term that attracts the contour towards strong image gradients (the true edge).

(d) Active contours are justified because they incorporate an explicit smoothness constraint alongside image-gradient attraction, allowing the contour to snap tightly to the true underlying boundary while suppressing small, noise-induced irregularities — producing a refined, accurate, and smooth boundary that the original rough detection could not achieve alone.

(e) The response follows Issue → Challenges → Refinement method → Reasoning, presented clearly.

26. Frequency-Based Enhancement

An image requires enhancement using frequency domain methods.

(a) Frequency-domain processing represents the image in terms of its constituent spatial frequencies (via the Fourier Transform) rather than as raw pixel intensities. Enhancement in this domain involves selectively amplifying or suppressing specific frequency bands — for example, boosting high frequencies to sharpen detail or suppressing low frequencies to correct illumination — before transforming back to the spatial domain.

(b) Key issues with spatial domain methods for this task include: (i) spatial convolution-based filters can be less precise or computationally efficient when a specific, well-defined frequency band needs to be targeted; and (ii) global effects such as periodic noise patterns or broad illumination variation are more naturally represented and isolated in the frequency domain than via local spatial operations.

(c) A frequency-domain high-boost (or high-pass) filter is applied: the image is transformed via the FFT, an appropriate filter (attenuating low frequencies while retaining/boosting high frequencies) is applied to the frequency spectrum, and the inverse FFT yields the enhanced image.

(d) The frequency-domain technique is justified because it provides direct, explicit control over which frequency components are amplified or suppressed, which is especially effective for global or periodic patterns that are not easily isolated by a local spatial convolution kernel of practical size.

(e) The response follows Requirement → Spatial-domain limitations → Frequency-domain method → Reasoning, presented clearly.

27. Mixed Noise Scenario

An image contains both Gaussian and impulse noise.

(a) This scenario presents a compounded challenge because two noise types with fundamentally different statistical characteristics are present simultaneously: Gaussian noise affects every pixel with small random deviations, while impulse (salt-and-pepper) noise corrupts a smaller fraction of pixels entirely, replacing them with extreme values.

(b) Key issues affecting filtering include: (i) a median filter, while effective against impulse noise, does not optimally reduce the additive Gaussian component; and (ii) a Gaussian/mean filter, while effective against additive noise, does not remove impulse spikes and can even spread them into neighboring pixels.

(c) A combined sequential approach is applied: a median filter is applied first to remove the impulse noise component, followed by a Gaussian smoothing filter to reduce the remaining additive Gaussian noise. Alternatively, an adaptive hybrid filter (e.g., a switching median filter combined with Gaussian smoothing) can be used.

(d) Applying the median filter first is justified because it eliminates the extreme impulse values before any averaging-based operation is performed; if the Gaussian filter were applied first, it would blend the impulse spikes into surrounding pixels, making them harder to remove afterward. The subsequent Gaussian filter then addresses the remaining, purely additive noise without impulse interference.

(e) The response follows Mixed-noise challenge → Filtering issues → Combined method → Reasoning for order, presented clearly.

28. Edge Preservation Filtering

A filtering method is required to remove noise but preserve edges.

(a) This scenario requires a filter that can simultaneously suppress unwanted noise in homogeneous regions while explicitly avoiding the blurring of true edges — a requirement that standard linear smoothing filters (mean, Gaussian) cannot satisfy, since they smooth indiscriminately across both.

(b) Key issues in filtering here include: (i) linear filters average across intensity discontinuities, blurring edges; (ii) the difficulty of distinguishing a true edge from a large, localized noise fluctuation using intensity information alone; and (iii) balancing the degree of smoothing against edge fidelity.

(c) A bilateral filter is applied. It computes a weighted average of neighboring pixels where the weight depends on both spatial distance (as in a standard Gaussian filter) and intensity similarity (a range/photometric term), so that only pixels with similar intensity to the center pixel contribute significantly to the smoothing.

(d) The bilateral filter is justified because its dual (spatial + range) weighting means that pixels across a strong edge — where intensity differs greatly — contribute little to the average, thus preserving the edge, while pixels within a homogeneous region — where intensities are similar — are smoothed together to remove noise, directly satisfying the combined requirement.

(e) The response follows Requirement → Filtering issues → Technique → Reasoning, presented clearly.

29. Region Growing Failure

Region growing fails due to incorrect seed selection.

(a) Region growing segments an image by starting from one or more seed pixels and iteratively adding neighboring pixels that satisfy a similarity criterion. If a seed is placed on a noisy, atypical, or boundary pixel rather than within a representative, homogeneous area of the intended region, the growth process can produce an incorrect, incomplete, or overly large/small region.

(b) Key challenges in region growing here include: (i) high sensitivity of the final result to the exact location of the initial seed; (ii) risk of a seed landing on noise or near a boundary, causing erroneous propagation; and (iii) the need for representative seed placement to obtain a correct result.

(c) An improved, automatic seed-selection strategy is applied — for example, selecting seeds from local intensity maxima/minima within homogeneous interior areas identified via histogram or local-variance analysis, or using multiple candidate seeds and selecting those consistent with the majority result.

(d) Automatic, statistically informed seed selection is justified because it removes the reliance on a single, possibly poorly chosen manual seed; by choosing seeds from regions verified to be homogeneous and representative (e.g., low local variance, consistent with the dominant intensity of the intended region), the region-growing process becomes far less sensitive to the specific seed location and produces a more reliable result.

(e) The response follows Cause of failure → Challenges → Correction → Reasoning, presented clearly.

30. Industrial Inspection Enhancement

An industrial system requires detecting defects in noisy images.

(a) This scenario defines a combined, multi-stage requirement: the system must remove noise from the captured images (to avoid false defect detections), preserve or even enhance the edges corresponding to true defects (so they remain detectable), and then reliably segment/isolate the defect regions for automated inspection.

(b) Key issues in detection here include: (i) noise potentially being mistaken for defects (false positives) or masking true defects (false negatives); (ii) the trade-off between noise suppression and edge/defect preservation; and (iii) the need for a segmentation step that can reliably isolate true defect regions from the background after enhancement.

(c) An enhancement-and-segmentation pipeline is applied: (1) a bilateral or median filter is used to denoise the image while preserving edges; (2) an edge-preserving enhancement or the Canny edge detector highlights defect boundaries; (3) thresholding (e.g., Otsu's method) or region-based segmentation isolates the defect regions; and (4) morphological operations (opening/closing) clean up the final result by removing residual small artifacts.

(d) This sequence is justified because each stage builds logically on the previous one: denoising first prevents noise from corrupting later edge/segmentation steps; using an edge-preserving filter (rather than a generic smoother) ensures the defect boundaries needed for detection are not lost; segmentation then reliably isolates candidate defect regions from the now-cleaner image; and the final morphological clean-up removes any small, spurious artifacts that survived the earlier stages, yielding a robust end-to-end pipeline suited to automated industrial inspection.

(e) The response follows Requirement → Detection issues → Pipeline → Reasoning for the sequence, presented in a clear, structured manner.

Code of Conduct:

I, V M SAI BHARGAV (1923240126), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V M Sai Bhargav

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand